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ORIGINAL ARTICLE

A deep learning-driven method for safe and effective ERCP cannulation

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Abstract

Purpose In recent years, the detection of the duodenal papilla and surgical cannula has become a critical task in computer-assisted endoscopic retrograde cholangiopancreatography (ERCP) cannulation operations. The complex surgical anatomy, coupled with the small size of the duodenal papillary orifice and its high similarity to the background, poses significant challenges to effective computer-assisted cannulation. To address these challenges, we present a deep learning-driven graphical user interface (GUI) to assist ERCP cannulation.

Methods Considering the characteristics of the ERCP scenario, we propose a deep learning method for duodenal papilla and surgical cannula detection, utilizing four swin transformer decoupled heads (4STDH). Four different prediction heads are employed to detect objects of different sizes. Subsequently, we integrate the swin transformer module to identify attention regions to explore prediction potential deeply. Moreover, we decouple the classification and regression networks, significantly improving the model's accuracy and robustness through the separation prediction. Simultaneously, we introduce a dataset on papilla and cannula (DPAC), consisting of 1840 annotated endoscopic images, which will be publicly available. We integrated 4STDH and several state-of-the-art methods into the GUI and compared them.

Results On the DPAC dataset, 4STDH outperforms state-of-the-art methods with an mAP of 93.2% and superior generalization performance. Additionally, the GUI provides real-time positions of the papilla and cannula, along with the planar distance and direction required for the cannula to reach the cannulation position.

Conclusion We validate the GUI's performance in human gastrointestinal endoscopic videos, showing deep learning's potential to enhance the safety and efficiency of clinical ERCP cannulation.

Keywords Deep learning · Duodenal papilla · Real-time detection · Computer-assisted cannulation · ERCP

Introduction

Due to factors such as improved living standards and an aging population, the incidence of biliary and pancreatic diseases, such as biliary stones, biliary tumors, and pancreatic tumors, has been on the rise [1]. These conditions may lead to uncomfortable symptoms such as jaundice and abdominal pain, and in severe cases, they can even be life-threatening [2]. Endoscopic retrograde cholangiopancreatography (ERCP) is internationally recognized as the gold standard for the diagnosis of pancreaticobiliary diseases. ERCP is a technique in which an endoscope is advanced into the descending part of the duodenum, the duodenal papilla is identified, and surgical instruments and guidewires are cannulated through a surgical cannula into the orifice of the papilla (see Fig. 1, this critical process establishes the working channel for all subsequent diagnosis and treatment [3]). Subsequently, contrast agents are injected, and X-ray images are taken to visualize the pancreatic and bile ducts [4].

ERCP is often referred to as the crown jewel of digestive endoscopic surgery. The ERCP process involves complex operations such as crossing different lumens and precise cannulation in a dynamic constricted environment, which significantly increases the difficulty of intervention and surgical risks, and places a heavy reliance on the skills and experience of the physician. The success rate of cannulation is reported to be only 86% according to research [5], and in cases of initial cannulation in patients, the success rate drops further to 79.5% due to the papilla occlusion and contraction.

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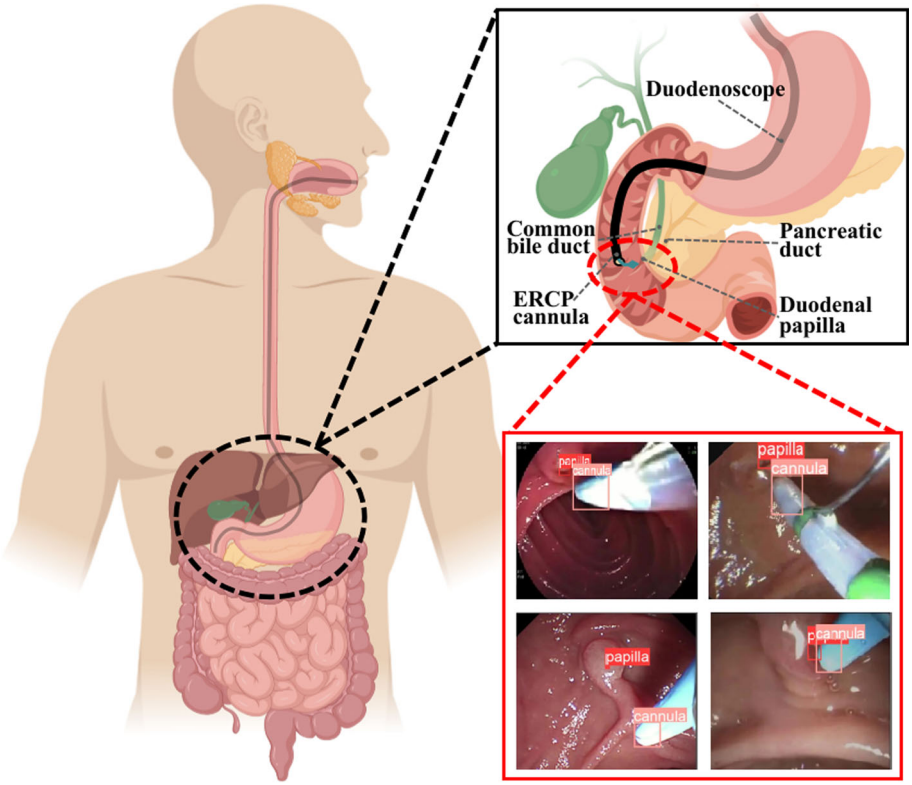


Fig. 1 Schematic representation of ERCP surgery. The inset within the black frame provides a magnified illustration of the cannulation site during the ERCP. The content within the red frame depicts clinical examples of ERCP surgeries in different individuals, including the duodenal papilla region

Researchers anticipate that computer-assisted surgical techniques based on deep learning will enhance the flexibility, accuracy, and stability of ERCP procedures [6, 7].

The application of computer-assisted technology in human cavity surgery has undergone more than a decade of development. He et al. [8] have provided reliable methods for the assisted intervention of endoscopic upper gastrointestinal procedures based on deep learning. Nonetheless, they did not address the issue of precise cannulation in ERCP. Jiang et al. [6] have proposed a method that combines Faster-Rcnn [9] and Median Flow [10] for duodenal papilla detection and automatic tracking of the surgical cannula. Although this approach offers a novel perspective on duodenal papilla localization and the extraction of the surgical cannula intervention direction in computer-assisted cannulation procedures, it is relatively complex, demonstrates lower detection accuracy, and lacks a visual interface. Consequently, overall safety considerations are compromised. The pancreaticobiliary area is located deep within the digestive tract, with a narrow field of view and limited operating space. Additionally, the small duodenal papillary orifice, with its high similarity to the surrounding background, makes ERCP cannulation driven by existing deep learning methods an exceptionally challenging task.

In this paper, we focus on the key tasks of computer-assisted technology to accomplish precise cannulation: duodenal papilla localization and surgical cannula localization as well as guidance interface design, providing insights into guiding ERCP cannulation operations safely and efficiently. Recent years have witnessed significant progress in object detection tasks using convolutional neural networks [11–13]. Simultaneously, these advancements have further stimulated the development of deep learning methods in the field of gastrointestinal endoscopy. [14, 15]. In light of these developments, we introduce a deep learning-based method to detect the duodenal papilla and surgical cannula to obtain their positions. This method faces two primary challenges that must be addressed. First, datasets play a critical role in training neural network models. Notable benchmark datasets, such as COCO [16], have significantly advanced the field of object detection. However, to date, no publicly available datasets are specifically dedicated to the detection of duodenal papillae and surgical cannulas. Therefore, under the guidance of clinical experts, we introduce the papilla and cannula (DPAC) dataset for the first time, designed to drive neural network models. Second, most previous deep convolutional neural networks have been primarily optimized for natural scene images, with low generalization performance in ERCP cannulation tasks. In contrast, the task of detecting duodenal papillae and surgical cannulas is particularly challenging due to the high similarity in color between the endoscopic image background and the duodenal papilla, the uncertain positioning of the duodenal papilla (as objects in datasets like COCO are generally centered), and the significant size disparity between

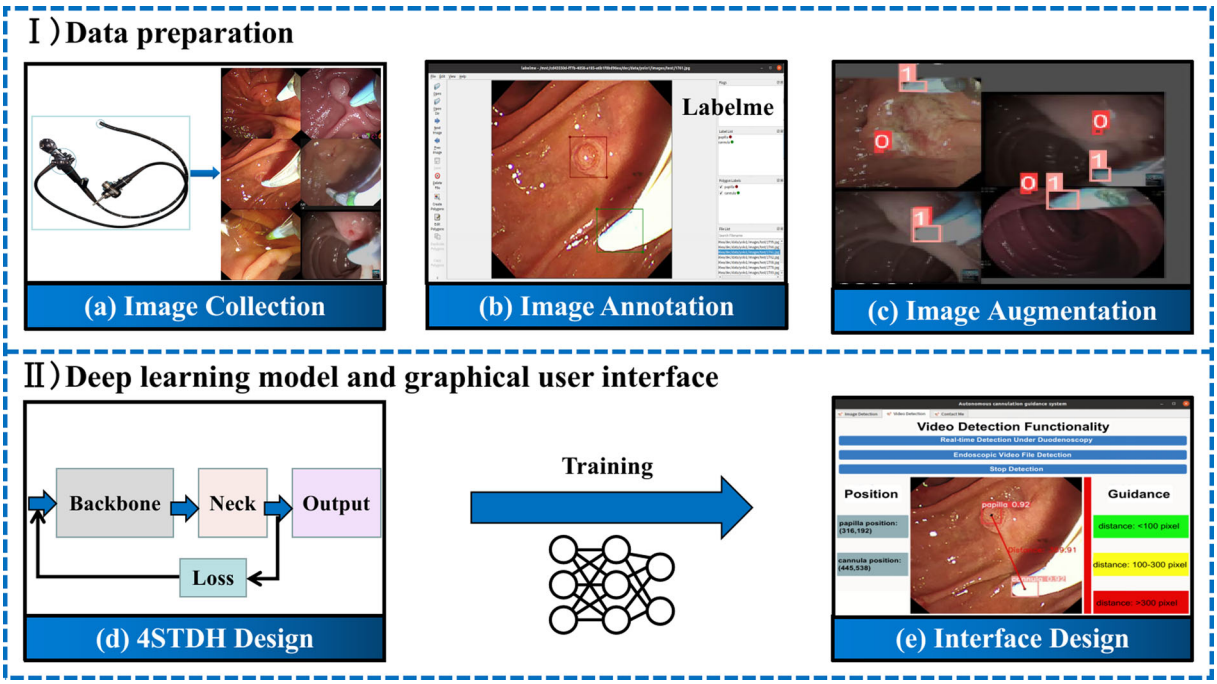


Fig. 2 Overall workflow of the cannulation guidance system during ERCP

the duodenal papilla and the surgical cannula. Hence, for the specialized surgical scenario of ERCP cannulation, we have introduced a deep learning method for duodenal papilla and surgical cannula detection, utilizing four swin transformer decoupled heads (4STDH). Our main contributions can be summarized as follows:

(i) We have established the first publicly available dataset specifically designed for ERCP cannulation guidance, named DPAC. This dataset has been annotated under the guidance of the clinical physician to label the positions and categories of the duodenal papilla and the surgical cannula.
(ii) We propose a novel framework, namely 4STDH. Our 4STDH leverages four different prediction heads, the classification and regression feature decoupling, and the swin transformer to solve the problems of large differences in object size, uncertain object positions, and the high similarity between the duodenal papilla and the background, respectively.
(iii) We have designed a real-time guidance interface for ERCP cannulation driven by deep learning methods, which intuitively displays the location and distance information of the duodenal papilla and surgical cannula, and provides cannulation warning information. The code is available at https://github.com/yuyingliu-1/4STDH.

Method

Overview of the working pipeline

In this paper, we propose a deep learning-driven approach for detecting the duodenal papilla and surgical cannula within an assisted cannulation guidance system for ERCP. The overall workflow of the cannulation guidance system, utilizing the 4STDH detection model, is illustrated in Fig. 2. First, we collected a dataset of endoscopic images related to ERCP cannulation surgery and annotated it to create the DPAC training dataset for the 4STDH model. Next, the DPAC training set was augmented using techniques such as Mixup and Mosaic. The 4STDH framework was then trained on this augmented dataset to generate the detection model. Finally, a cannulation guidance interface was developed based on the 4STDH detection model to enhance the safety and efficiency of cannulation tasks in endoscopic videos or real-time endoscopy.

Dataset collection, annotation, and augmentation

After frame splitting and filtering of ERCP video clips (after the endoscope reaches the duodenum) from both Tianjin Medical University General Hospital and publicly available sources [17–19], we ultimately compiled a dataset of 1840 endoscopic images (from 12 patients), each featuring both the duodenal papilla and the surgical cannula. The images from individual patients exhibit a high level of specificity, as illustrated in Fig. 2a.

Subsequently, under the guidance of clinical experts, we utilized the Labelme tool to annotate the position and category of the duodenal papilla and the surgical cannula in the endoscopic images (as shown in Fig. 2b). As depicted in Fig. 3a, there is a degree of uncertainty in the object's positions within the DPAC dataset. According to Fig. 3b, the DPAC dataset displays considerable variation in object sizes, with the overall sizes of the objects being relatively small, thereby focusing on the detection of small objects. Therefore, based on these analyses of the dataset, we optimized the sizes of prior anchor boxes.

Due to the inherent specificity in each patient's endoscopic images, we applied data augmentation (as shown in Fig. 2c) to augment the dataset before training the 4STDH detection model. This augmentation aims to enhance the model's robustness to the images captured from different environments. Notably, we employed MixUp and Mosaic techniques.

Framework of the 4STDH

Object detectors based on deep learning can be divided into many types: single-stage detectors, two-stage detectors, anchor-based detectors, and anchor-free detectors. Given the outstanding performance of Yolov5 [12] in the field of object detection, we designate it as our baseline. The 4STDH model framework we designed also consists of three parts: the backbone, the neck, and the prediction head (output) (Fig. 4). We will introduce these three parts of the structure in detail in the following content.

Backbone

The backbone network is primarily used to extract features from images. Backbone networks have robust feature extraction capabilities in classification and other tasks. Typically, researchers only need to fine-tune the backbone to better suit specific tasks. In this work, we adopt CSPDarknet53 [20] as the backbone.

Neck

The design of the neck aims to further explore the image features extracted by the backbone, reprocess, and effectively utilize the feature maps extracted at different stages. We have integrated the swin transformer module [21] for superior performance. Through the design advantages of hierarchical feature representation and shifted windows, swin transformer can capture features of different scales of images and effectively model long-distance dependencies.

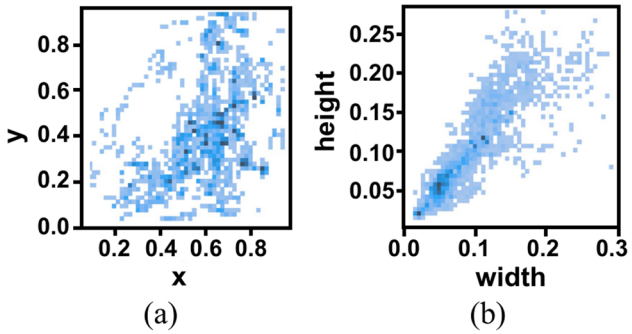


Fig. 3 Bounding boxes of labeled datasets. **a** is the position of the center point relative to the entire picture; **b** is the height-to-width ratio of the target in the picture relative to the entire picture

Prediction head

The prediction head is designed to utilize the feature maps extracted by the backbone and neck to further detect the position and category of objects. The structure of the four prediction heads we designed can better handle scenes with large differences in object sizes and an overall small scale. As shown in Fig. 4, our additional prediction head is generated from low-level, high-resolution feature maps, making it more sensitive to small objects.

The introduction of decoupled heads is to address the conflict between classification and regression tasks. Similarly, in the head of 4STDH, multiple tasks are set, including classification, regression, and detection. To improve the accuracy of the algorithm, it is necessary to decouple these tasks, dividing the prediction head into two independent branches: one for classification tasks and the other for object detection tasks.

Loss function

The 4STDH detection model in this study contains three loss functions: the classification loss L_{cls}, which measures the accuracy of the model's predictions for object classes; the bounding box regression loss L_{box}, which evaluates the deviation between predicted bounding boxes and ground truth boxes; and the confidence loss L_{conf}, which assesses the model's ability to accurately determine whether a bounding box contains an object. The L_{box} is calculated using the CIoU function [22]. The L_{conf} and the L_{cls} are calculated using the binary cross-entropy loss function (BCEWithLogits Loss). The overall loss function is defined as:

$$L = \lambda_1 L_{box} + \lambda_2 L_{conf} + \lambda_3 L_{cls} \quad (1)$$

As a rule of thumb, we set the default coefficient weights λ_1, λ_2, λ_3 to 0.05, 1.0, and 0.5, respectively.

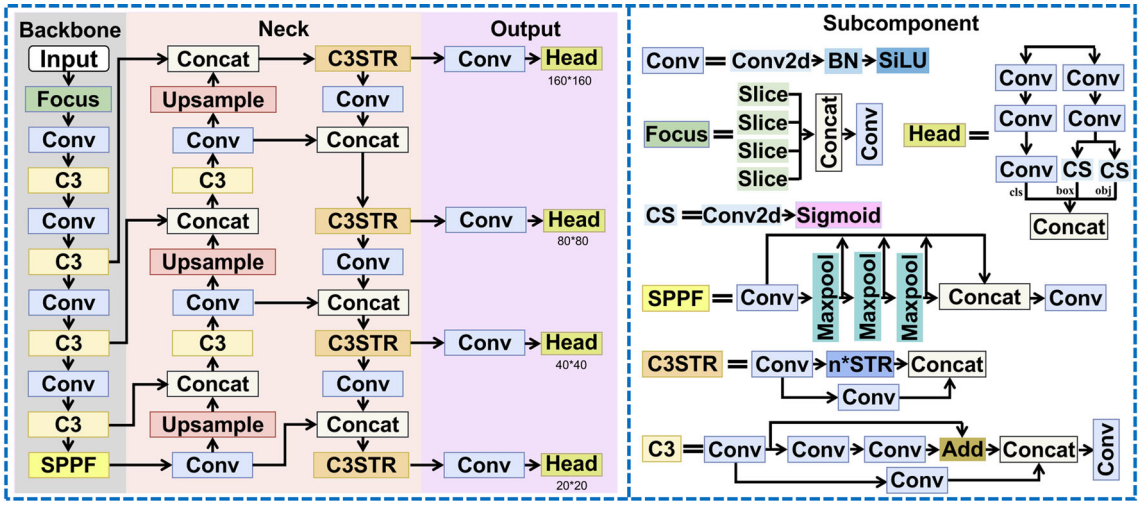


Fig. 4 Architecture of the 4STDH

Experiments and results

Experimental setup

We implemented our 4STDH detection models in the Ubuntu 20.04.3 LTS operating system using the pytorch 1.7.0+cu110 framework based on python 3.6 programming language, CUDA 11.0, and CUDNN 8.0. The networks used the NVIDIA GeForce RTX 3080 GPU as the accelerator for training. The DPAC dataset is divided into a training set, a validation set, and a test set in a ratio of 7:1:2, and then, the model training is performed after data augmentation. Our models are trained for 300 epochs using SGD, with a batch size of 8, and a learning rate of 10^{-2}.

Evaluation metrics

To comprehensively evaluate the 4STDH model's performance, we adopted multiple evaluation metrics, including precision (P), recall (R), average precision (AP), mean average precision (mAP), and F1 score for the model's overall performance. The calculation formulas are as follows:

$$P = \frac{TP}{TP + FP} \tag{2}$$

$$R = \frac{TP}{TP + FN} \tag{3}$$

$$AP = \int_0^1 P(R)dR \tag{4}$$

$$mAP = \frac{1}{C}\sum_{n=1}^{C} AP_n \tag{5}$$

$$F_1 = 2 \times \frac{P \cdot R}{P + R} \tag{6}$$

where TP is the number of correctly detected positive samples, FP is the number of negative samples that are incorrectly detected as positive samples, FN is the number of positive samples that are incorrectly detected as negative samples, and C is the number of categories.

Comparisons with the state-of-the-art

To quantitatively evaluate the performance of the proposed 4STDH algorithm, a comparative analysis was performed with several state-of-the-art algorithms. We selected five state-of-the-art methods, Faster-Rcnn (two-stage) [9], SSD (one-stage) [11], CenterNet (anchor-free) [23], Yolov5s (anchor-based) [12], and Yolov8s (anchor-free) [13] to compare with our method in comparative experiments on the DPAC dataset. From the quantitative performance evaluation results in Table 1, it can be observed that our 4STDH performs the best, especially with papilla and cannula achieving $AP_{0.5}$ of 87.14% and 99.44%, respectively. Even the $mAP_{0.5}$ reaches 93.29%, surpassing other algorithms. It is noteworthy that the newly proposed Yolov8s model didn't perform as expected in the detection of duodenal papilla and surgical cannula, and was even inferior to the performance of Yolov5s. We selected representative images to display the detection test results. Figure 5 displays the detection results for large objects, small objects, overlapping objects, and images where objects are similar to the background. And the 4STDH detection algorithm demonstrates excellent detection performance.

The F1 score is an important metric for classification, representing the harmonic mean of precision and recall. Its value ranges between 0 and 1, with higher values indicating better performance. From Fig. 6, it can be observed that the F1 curve of the 4STDH algorithm is "wide," indicating a pretty

Table 1 Comparison of our method to different methods on the DPAC dataset

Method	Unit (%)	P	R	$AP_{0.5}$	$mAP_{0.5}$	$mAP_{0.9}$
Faster-Rcnn (Restnet50)	Papilla	35.82	77.01	58.86	79.15	43.27
	Cannula	89.71	99.46	99.43		
SSD	Papilla	36.14	80.27	68.29	79.50	43.72
	Cannula	82.69	98.65	90.70		
CenterNet (Hourglass)	Papilla	87.80	30.17	52.04	73.19	42.79
	Cannula	97.30	61.02	94.33		
Yolov5s	Papilla	89.20	85.22	86.41	92.67	52.33
	Cannula	97.71	98.33	98.93		
Yolov8s	Papilla	83.41	68.50	71.36	83.60	45.36
	Cannula	96.32	97.11	95.84		
4STDH	Papilla	89.21	85.72	**87.14**	**93.29**	**52.70**
	Cannula	98.62	99.03	**99.44**		

The bolded values are the best results in each metric

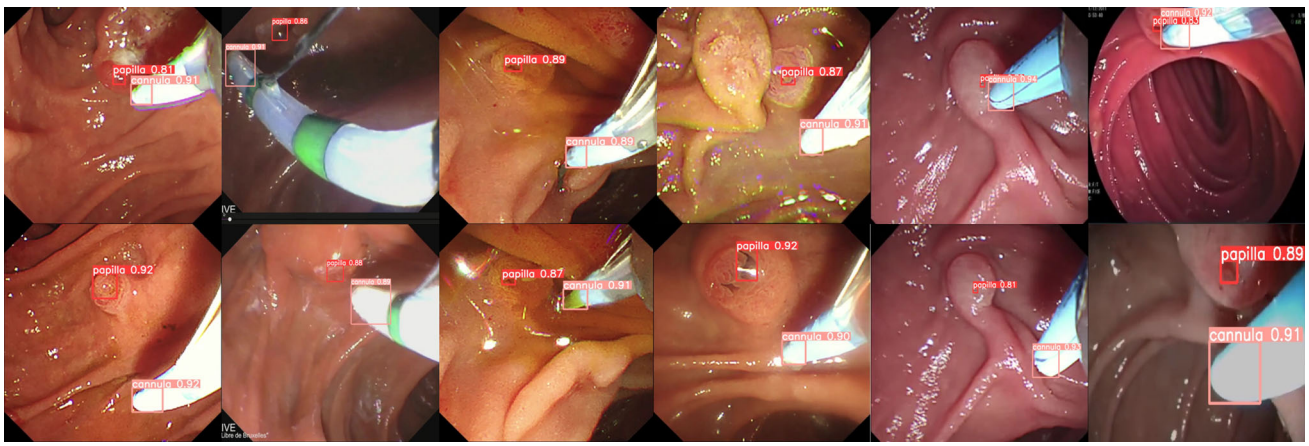


Fig. 5 Some visualization results from our 4STDH on DPAC dataset, different category use bounding boxes with different color

large confidence threshold range where it performs well in both recall and precision.

Ablation study

To verify the contribution of the four prediction heads, the swin transformer module, and the classification and regression task decoupling head integrated into the 4STDH, we conducted an ablation study based on fivefold cross-validation with Yolov5s as the baseline. The specific results are presented in Table 2, where Baseline+4 H represents the addition of the four prediction heads, Baseline+CBAM represents the addition of the CBAM attention module, Baseline+T represents the addition of the transformer module, Baseline+ST represents the addition of the swin transformer module, and Baseline+DH represents the addition of the decoupling head. As shown in Table 2, the addition of CBAM [24] and transformer [25] decreased some evaluation indicators of the model, but the addition of the swin transformer module improved the overall performance of the model. The

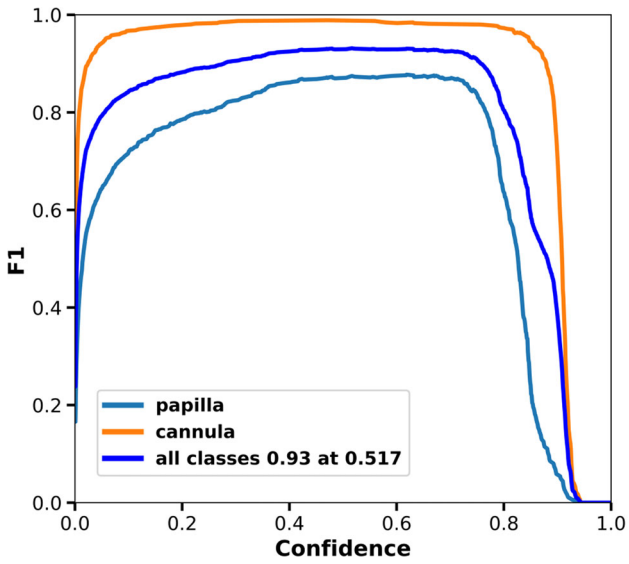


Fig. 6 Relationship between F1 score and confidence threshold

Table 2 Ablation study on the DPAC dataset

Category	Model	1st fold	2nd fold	3rd fold	4th fold	5th fold	Means
Papilla	Baseline	86.42	72.76	79.23	86.40	86.44	82.55
	Baseline+4 H	86.98	72.82	80.20	86.98	87.11	82.82
	Baseline+CBAM	86.12	72.74	79.24	86.11	86.50	82.14
	Baseline+T	86.46	73.10	79.33	86.42	86.49	82.36
	Baseline+ST	86.70	74.53	79.34	86.73	86.59	82.78
	Baseline+DH	87.21	74.97	81.62	87.12	87.20	83.62
	4STDH	87.21	85.66	86.71	87.15	87.22	**86.79**
Cannula	Baseline	98.90	95.89	95.60	98.89	98.90	97.64
	Baseline+4 H	98.96	95.88	95.60	98.97	98.90	97.66
	Baseline+CBAM	98.92	95.91	95.61	98.67	98.70	97.56
	Baseline+T	98.99	96.02	95.62	99.01	99.05	97.74
	Baseline+ST	99.13	96.91	96.77	99.11	99.11	98.21
	Baseline+DH	99.40	96.89	98.63	99.40	99.39	98.74
	4STDH	99.42	97.71	98.88	99.43	99.43	**98.97**

The bold values are the best Means of all detection models for papilla and cannula, respectively

results of Baseline+4 H and Baseline+DH show that the addition of the small objects prediction heads and decoupling heads improves the model's performance in all aspects to a certain extent, especially in the category of duodenal papilla. After adding additional detection heads, although the computational and memory costs increase, the performance of small object detection becomes better, which is worthwhile. Our approach, which integrates all these techniques, leads to substantial performance gains. The fivefold cross-validation results indicate that the performance of the models is similar in the first, fourth, and fifth folds. However, in the second and third folds, our 4STDH model significantly outperforms the baseline.

Heatmaps are valuable tools for observing the degree of attention the model assigns to each pixel or region within an image. Warm-toned areas indicate regions that the model considers more likely to contain objects, while cold-toned areas represent regions of lesser importance. By analyzing heatmaps, we gain deeper insights into the model's decision-making process and can assess its ability to accurately localize objects. In this study, we employed the Gradient-weighted Class Activation Mapping (Grad-CAM) technique to generate heatmaps. By calculating the gradient of the predicted class score and the feature map of the last convolutional layer, we determine the importance of each feature map to a specific class. To illustrate the impact of different attention mechanisms, we visualize the heatmaps generated by the Baseline, Baseline+CBAM, Baseline+T, and Baseline+ST models, as shown in Fig. 7. From the figure, it is evident that the models incorporating attention modules are able to focus more purposefully on the objects (i.e., the duodenal papilla and the surgical cannula) compared to the baseline model without attention mechanisms. Furthermore, by comparing the heatmaps generated by different attention modules (i.e., the 3rd, 4th, and 5th columns), we observe a trend consistent with the findings in Table 2. Specifically, when compared to the CBAM and transformer modules, the swin transformer module enables the model to focus more effectively on the object regions, thereby extracting more relevant and important feature information.

Interface design

To enhance the visualization performance of computer-assisted ERCP cannulation systems, we have designed an interface (Fig. 8) for the 4STDH detection model, which can improve the safety and efficiency of the surgery. We implemented a graphical user interface (GUI) using PyQt5 and OpenCV libraries. We utilized multithreading to handle real-time video detection, preventing the interface from being blocked during the detection process. In this interface, you can choose whether to input a video stream or a real-time endoscopic acquisition image by clicking a button. The guidance interface can respond to commands in real-time, demonstrating good operational stability and high accuracy.

The guidance interface provides detailed positional information, classification, and prediction accuracy for both the duodenal papilla and the surgical cannula. Additionally, it indicates the distance and directional requirements for aligning the surgical cannula to the target cannulation position. To enhance safety and operational precision, the computer-assisted ERCP cannulation guidance interface includes a color-coded warning system based on distance. When the distance is less than 100 pixels, the surgical cannula is deemed to be very close to the duodenal papilla orifice, signaling readiness for cannulation with a green warning. For distances

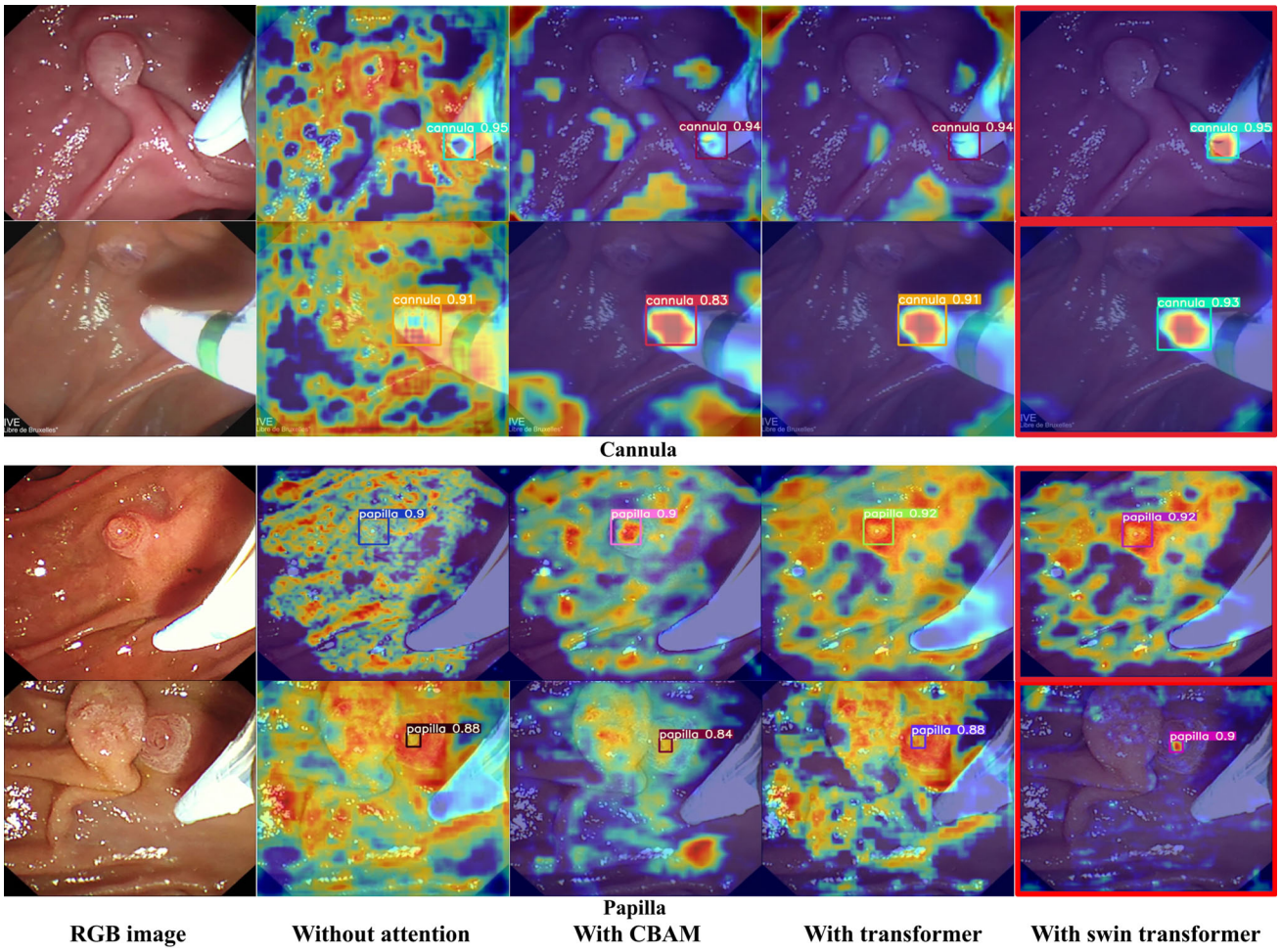


Fig. 7 Output heatmap obtained from models without attention and with different attention modules

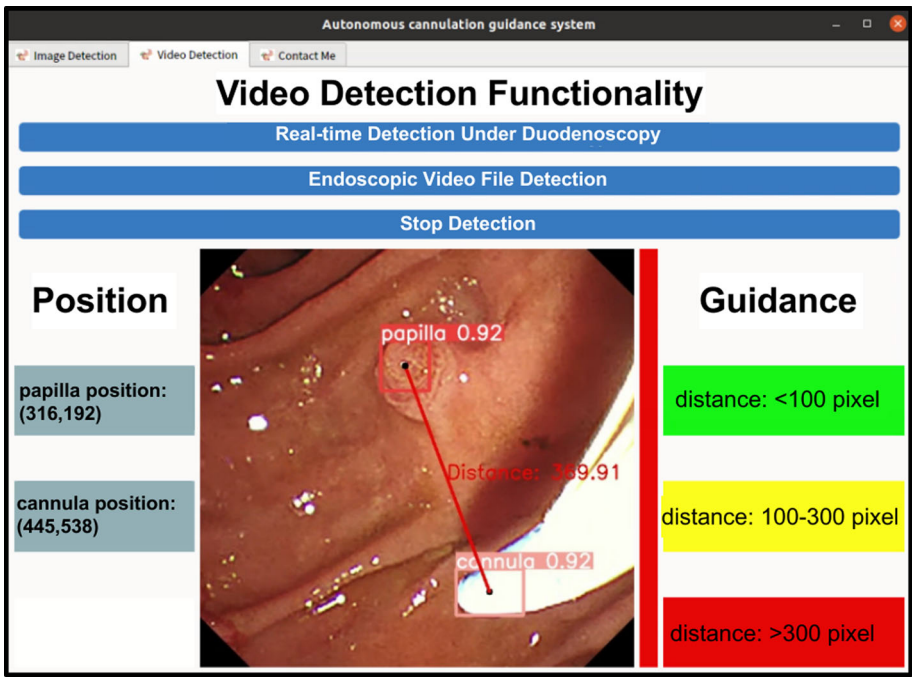


Fig. 8 Guidance interface with the 4STDH for computer-assisted ERCP cannulation

between 100 and 300 pixels, the system indicates moderate proximity, suggesting incremental adjustments to locate the papilla, marked by a yellow warning. When the distance exceeds 300 pixels, indicating a substantial gap between the surgical cannula and the duodenal papilla, the system prompts a more extensive movement to locate the papilla, accompanied by a red warning.

Discussion

This study presents a comprehensive pipeline encompassing dataset creation, augmentation, network model construction, and GUI design, culminating in a complete algorithmic solution. The 4STDH model is a single-stage object detection model designed to achieve real-time performance (the interactive interface achieves a response frame rate of approximately 32 fps in clinical applications) while maintaining accuracy. Studies report that the success rate for physician-led cannulation during a patient's first ERCP attempt is only 79.5% [5]. Tailored specifically to assist ERCP cannulation, the algorithm provides intuitive intraoperative guidance to physicians, helping to save procedure time and improve cannulation success rates. It also advances technology for robotic surgical navigation and serves as a valuable training tool for novice physicians.

The annotated DPAC dataset we introduced provides a foundation for training state-of-the-art deep learning-based methods (Table 1), facilitating research in related domains. Experimental results indicate that, although other deep learning methods can also accomplish the task and the Yolov5 model shows considerable accuracy, the 4STDH method exhibits the overall best performance in the application of ERCP cannulation scenarios (among the other methods). Specifically, our method increases the $AP_{0.5}$ metric for papilla detection from 58.86% with Faster-Rcnn to 87.14%. Notably, the Yolov8s model underperforms relative to the Yolov5s model in this task, underscoring the variability of model performance across different application scenarios. These findings highlight the importance of designing and selecting models tailored to the specific characteristics of each application.

Ablation experiments using fivefold cross-validation (Table 2) show that, compared to the Baseline, our 4STDH model achieved an average increase of 4.24% in $AP_{0.5}$ for papilla detection and 1.33% for cannula detection. A comparison of values in Table 2 further demonstrates that our four decoupled prediction heads are effective at detecting objects with significant size variation and perform exceptionally well in identifying very small instances. Furthermore, the swin transformer module enhances the model's ability to distinguish between background and object, especially in challenging scenarios where the duodenal papilla is highly similar to the background. Notably, in the fivefold cross-validation study, the 4STDH model consistently achieved an $AP_{0.5}$ of over 85% for papilla detection, whereas the Baseline model showed considerable fluctuation. In the second and third fold experiments, the $AP_{0.5}$ for papilla detection increased by 12.9% and 7.48%, respectively, while in the first, fourth, and fifth folds, the model evaluation scores remained similar. This demonstrates the high stability and strong generalization ability of the 4STDH algorithm, while also confirming the importance of designing and selecting suitable detection techniques based on the characteristics of the detection scenario.

We integrated the 4STDH model (or other advanced models) into the interface of a computer-assisted ERCP cannulation guidance system and validated its performance using ERCP cannulation endoscopic videos captured in various human gastrointestinal environments. The interface provides real-time responses and delivers various visual information during validation. The interface is straightforward to integrate, allowing researchers to easily replace the driving model with one more suitable for other specific scenarios or upgrade to better algorithms in the future.

However, our method still has some limitations that need to be addressed in future work. Although our 4STDH model achieves high accuracy in cannula detection and surpasses some advanced algorithms in papilla detection, there remains room for further improvement in papilla detection accuracy. Hereafter, tiny object detection technology can be introduced to further improve the accuracy of papilla detection [26]. Additionally, as our method relies on annotated datasets, it could be enhanced by adopting semi-supervised or self-supervised learning approaches.

In future work, we plan to integrate the designed ERCP cannulation guidance interface with the surgical robot system to further realize automated intervention in ERCP procedures. We look forward to applying this integrated system in real-time clinical ERCP, enhancing the safety and efficiency of ERCP procedures and thereby reducing complications.

Conclusions

In this paper, we present a real-time guidance interface for computer-assisted ERCP cannulation, leveraging our proposed 4STDH detection algorithm. The 4STDH algorithm integrates four different prediction heads, the classification and regression feature decoupling, and the swin transformer module, which can effectively address challenges such as significant size differences in the duodenal papilla and the surgical cannula, similarity between the duodenal papilla and the background, and uncertainties in object positions. These improve both the accuracy and robustness of detection. Additionally, our designed guidance interface consistently

provides real-time and stable access to multiple visual information. In conclusion, our research results emphasize the positive impact of the proposed method in the field of computer-assisted ERCP cannulation, with the potential to improve the safety and efficiency of examination and treatment. It holds promise as a valuable tool for clinical practitioners performing ERCP procedures.

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Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

Ethical approval Approval was obtained from the ethics committee of Tianjin Medical University General Hospital (Ethics approval number:IRB2022-YX-046-01).

Informed consent Informed consent was obtained from all individual participants included in the study.

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